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Topic : Ports Concepts

So can we change the port numbers and how does the port number work .

- So first 8080 is our spring boot app port and second 3306 is our mysql database port

So types of ports in Docker :

```
ports:
  - "8080:8080"
```

- This format means : Host port : Container port
- In left side (8080) is our computer port where browsers uses this
- In right side (8080) is inside container port which the app runs here

Based on our current setup : MYSQL : Service :

```
ports:
  - "3307:3306"
```

- The MYSQL runs inside the container on 3306
- You can access from your PC using 3307
- So from host which is our laptop , we use localhost : 3307
- But from another container , we use mysql : 3306

Based on our spring boot app : In app service :

```
ports:
  - "8080:8080"
```

- The app is running inside the container on 8080
- You can access with browser as http://localhost :8080

Based on DB URL : In application.properties :

```
spring.datasource.url=jdbc:mysql://mysql:3306/electricity_db
```

- Inside docker network the mysql is the service name like hostname
- So 3306 is the container port used my mysql
- So not 3307 because the containers talks internally and not via ports host

So can we change ports : Yes we can change ports :

- By changing the app port in spring boot in app service in application.properties :

```
server.port=8080
```

- In application properties we have server port as 8080 and which means we should use this 8080 in docker app service

```
ports:
  - "8080:8080"
```

- So we can access with http://localhost:8080

So if we change the port in application.properties :

```
server.port=9090
```

We must also change to 9090 in : Docker : app service :

```
ports:
  - "9090:9090"
```

- Finally we can access with http://localhost:9090

By changing MYSQL host port in mysql service :

From this :

```
ports:
    - "3307:3306"
```

- Using this current port so from laptop we use local host as 3307
- So for app container we still use mysql : 3306 which is unchanged

To this :

```
ports:
    - "3308:3306"
```

- Using this new port in mysql service by using laptop we use local host as 3308
- But for app container we still use mysql : 3306 which is still unchanged

So when should you change the ports :

- Change port when port already in use like 8080 is busy then use 8081
- So with running applications use for app1 : 8080 and for app2 : 8081 to avoid conflicts

So in application.properties :

```
spring.datasource.url=jdbc:mysql://mysql:3306/electricity_db
```

- So when we use localhost as 3307 , this will fail inside docker and when we use mysql as 3306 , this will work
- So using local host as 3307 looks correct but its not
- So we think MYSQL is running on port 3307 , so should use local host : 3307 and this logic is only true from our laptop and not inside docker container
- So inside a container using localhost means the container itself and not our laptop and MYSQL is in a different container
- So the spring boot container tries localhost and looks inside and thus MYSQL is not found
- So we have spring boot container as application.properties and MYSQL container as mysql service in docker compose

So if spring boot says connect to localhost and this means :

- Connect to my spring boot with mysql and finally the connection will fail

So therefore :

- So when we use docker service name as mysql : 3306 in docker compose in mysql services , mysql becomes a hostname such as service inside docker network
- So spring boot uses mysql : 3306 and thus finds the mysql container present

Based on the real work flow :

- Spring boot container uses localhost : 3307 and finds MYSQL not found and thus connection will fail
- If spring boot container uses mysql : 3306 and thus finds MYSQL container there and connection works

```
ports:
  - "3307:3306"
```

- So 3307 is exists only for accessing MYSQL from your host machine for MYSQL Command Line

```
mysql -h localhost -p 3307
```